MILITARY SPECIFICATION

CABLE, ELECTRONIC, TOW

1. SCOPE

1.1 This specification covers electronic tow cable for submarine applications.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-572 — Cords, Yarns and Monfilaments-Organic Synthetic Fiber.
MIL-I-631 — Insulation, Electrical, Synthetic Resin Composition, Nonrigid.
MIL-C-12000 — Cable, Cord, and Wire, Electric, Packaging of.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B3 (1963) — Soft and Annealed Copper Wire.
ASTM B8 (1964) — Concentric-lay Stranded Copper Conductors, Hard, Medium Hard, or Soft.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 First article. - Prior to beginning production, a sample shall be tested as specified in 4.3.1 (see 6.2).

3.2. Material. - All material shall be as specified herein and shall be compatible with sea water environment. Any commercial material used by the contractor not covered by Government specifications, shall be approved by the command or agency concerned prior to its use in the end product. No material shall be plated to prevent corrosion.

3.3 Operating temperature range. - The cable shall be capable of meeting outboard environmental temperatures -54°C. to +65°C.

3.4 Lengths. - The cable shall be provided in lengths of 830 $^{+20}_{-0}$ feet or 1640 $^{+20}_{-0}$ feet or in multiples thereof as specified (see 6.1).

3.5 Design and construction. - The design and construction of the tow cable shall be as specified herein.

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3.5.1 Inner conductor. - The inner conductor shall consist of 7 strands of silver covered (40 microinches min.) soft annealed copper wire in concentric lay. Each strand shall be AWG 30 (0.010 inch diameter). Nominal overall diameter of the conductor shall be 0.33 inch. Conductor material shall be in accordance with ASTM-B3 and concentric-lay shall be in accordance with class B stranding of ASTM-B8.

3.5.2 The inner conductor shall be insulated with Teflon FEP, or equal, 0.054 $^{+0.001}_{-0.002}$ inch outside diameter. The maximum conductor eccentricity shall not exceed 0.005 inch (see 4.5.3.1).

3.5.3 Braid. - Braid shall be single braid AWG-38 silver covered (40 microinches minimum) copper wire. Overall diameter shall be 0.074 inch maximum.

Weave	2/2
Carriers	16
Ends	4
Picks/inch	21 ±10 percent

3.5.4 Jacket. - The jacket shall be of DuPont Zytel Nylon 33 or 37X, or equivalent. Outside diameter shall not exceed 0.082 inch.

3.5.5 The four coaxial cables constructed as specified in 3.5.1 through 3.5.4 shall be wound in a spiral with polyethylene fillers 0.033 inch diameter nominal and covered with a double wrap of 0.0005 minimum mylar tape in accordance with type G of MIL-I-631 with 50-percent overlap.

3.5.6 Cables shall be covered with a rayon braid approximately 0.005 inch thick impregnated with American Steel and Wire AMERTEX, or equivalent. The rayon shall be in accordance with MIL-C-572.

3.5.7 Armor. - Two layers of armor strands shall enclose the cable. The under layer shall be 18 strands with 2.00 ±0.200 inch right-hand lay. The outer layer shall have 24 strands with a 3.00 ±0.200 inch left-hand lay. There shall be no crossed-over armor strands in either lay. Each strand shall be 0.0375 inch diameter type 304 corrosion-resistant steel. The outside diameter of the complete cable shall be 0.350 ±0.005 inch. There shall be no welds, brazes, or splices in any of the armor strands. Each strand shall be smooth with no pitted surfaces and shall be free of scratches and dents. The tensile strength of any individual armor strand prior to forming shall be from 260,000 to 300,000 pounds for each square inch.

3.6 Cable core (Tank test). - The cable core shall be capable of withstanding 2,000 volts, root mean square (rms) 60 Hertz, for 5 minutes.

3.7 Hydrostatic pressure. - The cable shall withstand the hydrostatic pressure test specified in 4.5.2.1 without any sign of water penetration to the individual conductors.

3.8 Flexibility. - The cable shall withstand the flexibility test specified in 4.5.2.2 without sustaining any damage which would impair its serviceability.

3.9 Vibration. - The cable shall withstand the vibration test specified in 4.5.2.3 without sustaining any damage which would impair its serviceability.

3.10 Strength. - The cable shall have an ultimate strength of 9,500 pounds, when tested as specified in 4.5.2.4.

3.11 Insulation resistance. - The insulation resistance of the cable shall be not less than 40,000 megohms/1,000 feet, when tested as specified in 4.5.3.2.

3.12 Continuity. - Each conductor in the cable shall be continuous, when tested as specified in 4.5.3.3.

3.13 DC resistance. - The dc resistance of each conductor in the cable shall not exceed 0.0162 ohms per foot, when measured as specified in 4.5.3.4.

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3.14 Dielectric withstanding voltage. - The cable shall withstand 2,000 volts, 30 Hertz, ac for 1 minute without breakdown.

3.15 Marking. - The cable shall carry a marker tape under the first armor shield or under the mylar wrap. The tape shall be marked, at least every 24 inches, with the following:

(a) Manufacturer's name (or symbol designation)
(b) Year of manufacture (within 30 days)
(c) Specification number

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. - The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection. - First article inspection shall consist of the examination and tests specified in 4.5.

4.3.1 First article sample. - A 100 foot sample of cable constructed with specified materials and manufactured with equipment to be used in production shall be subjected to the first article inspection specified in 4.3. Unless otherwise instructed, the first article inspection shall be conducted at the Naval Ship Engineering Center, Philadephia Division, Submarine Antenna Quality Assurance Department (SAQAD), Code 6700D. The above sample shall be furnished to Code 6700D with complete test data of 4.5.1. A test report shall then be forwarded by SAQAD to the contracting officer and the contractor with an information copy to the Naval Electronic Systems Command, Code 05143. The manufacturer shall not continue production prior to receipt of approval by the contracting officer except at his own risk.

4.4 Quality conformance inspection. -

4.4.1 Production inspection. - Each production length of cable shall be subjected to the following tests:

Test	Paragraph no.
Insulation resistance	4.5.3.2
Continuity	4.5.3.3
Direct current resistance	4.5.3.4
Dielectric withstanding voltage	4.5.3.5

4.4.2 Production check inspection. - One specimen from every 33,000 feet of production cable shall be subjected to all tests specified in 4.5.

4.5 Test procedures. -

4.5.1 Physical examination. - One 15-foot specimen from every 8,000 feet of production cable shall be examined to determine conformance with the dimensional material and design and construction requirements specified herein.

4.5.2 Mechanical tests. -

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4.5.2.1 Hydrostatic pressure. - Hydrostatic pressure test shall be conducted by the use of water. A 10-foot length of cable shall be placed in a pressure tank with both ends extending from the test tank. A suitable fitting or packing gland shall be provided to insure that the water doesn't leak from around the outside of the cable. The cable shall then be subjected to the following test:

(a) Apply 900 psi, hydraulic pressure for 4 hours.
(b) Release pressure, but continue immersion for 15 to 18 hours.
(c) Apply 900 psi hydraulic pressure for 4 hours, then remove from tank.

After conducting hydrostatic pressure tests, examine cable for evidence of water. The cable shall then be subjected to 1000 volts, 60 Hertz between the conductors. The test voltage shall be applied for 1 minute. The test shall be performed at temperature of approximately 23°C. and maximum relative humidity of 60 percent.

4.5.2.2 Cable flexibility. - A length of cable shall be bent 180 degrees around each of two 14-inch in diameter steel pulleys as shown in figure 1. The pulleys shall not be coated with a soft plastic. With the cable under tension of 1,000 pounds, it shall be caused to move 4 feet in each direction over the pulleys 30,000 times, without sustaining any damage which would impair its serviceability. Breakage of any strand, loss of continuity, or displacement of any armor strand shall be considered a failure.

4.5.2.3 Vibration. - A length of the cable, assembled as shown on figure 2 shall be placed under a tension of 2700 pounds, using the method described in figure 3. Near its midpoint, the cable shall be vibrated at a frequency of 27 ±1 Hertz for each second (Hz), causing an amplitude of at least 4 inches at a point on the cable. The connecting rod from the vibrator to the cable shall be designed such that the cable does not break at the point of contact. The Gimbal required for this test shall be provided by the manufacturer. The duration of the vibration test shall be 20 hours.

4.5.2.4 Tension. - A 4-foot section cable with two suitable armor grips shall be subjected to a tensile test to determine conformance to 3.10.

4.5.3 Electrical tests. -

4.5.3.1 Cable cores. - All production lengths of insulated conductor shall be tank tested in a water tank containing a 5-percent wetting agent. A voltage of 2,000 volts rms, 60 Hertz shall be applied for 5 minutes.

4.5.3.2 Insulation resistance. - The insulation resistance of each coaxial cable shall be measured with a 100-volt direct current (dc) source at approximately 23°C and a maximum relative humidity of 60 percent. The insulation resistance of each cable shall be as specified in 3.11. The reading shall be taken after an elapsed time of 90 percent of the total charging time as required by the constants of the bridge used for measurements. No one value shall be less than 90 percent of any of the other three values measured in one assembly.

4.5.3.3 Continuity. - The continuity of each cable (inner and outer conductors) shall measured with a 6-volt, dc source and appropriate indicating devices.

4.5.3.4 DC resistance. - The dc resistance of each conductor (inner and outer) of each coaxial cable shall not exceed 0.0162 ohms for each foot when measured at a temperature of 23°C. and a relative humidity of 60 percent maximum.

4.5.3.5 Dielectric withstanding voltage. - Each coaxial conductor in the cable shall be tested with 2000 volts, 60 Hertz alternating-current (ac) for 1 minute between inner conductor and braid.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

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5.1 Preservation, packaging, packing and marking. - Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in *MIL-C-12000 (see 6)*.

6. NOTES

6.1 Ordering data. - *Procurement documents* should specify the following:

(a) Title, number, and date of this specification.
(b) Length required (see 3.4).
(c) Level of preservation, packaging, packing, and marking required (see 5.1).

6.2 First article. - Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples for those bidders offering a product which has been previously procured or tested by the Government and that bidders offering such products who wish to rely on such production or test must furnish evidence with the bid that prior Government approval *is presently appropriate* for the pending procurement. (See 4.3.1.)

6.3 Changes from previous issue. - The outside margins of this document have been marked "#" to indicate where changes (deletions, additions, etc.) from the *previous issue have been made*. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - EC

(Project 6145-N081)

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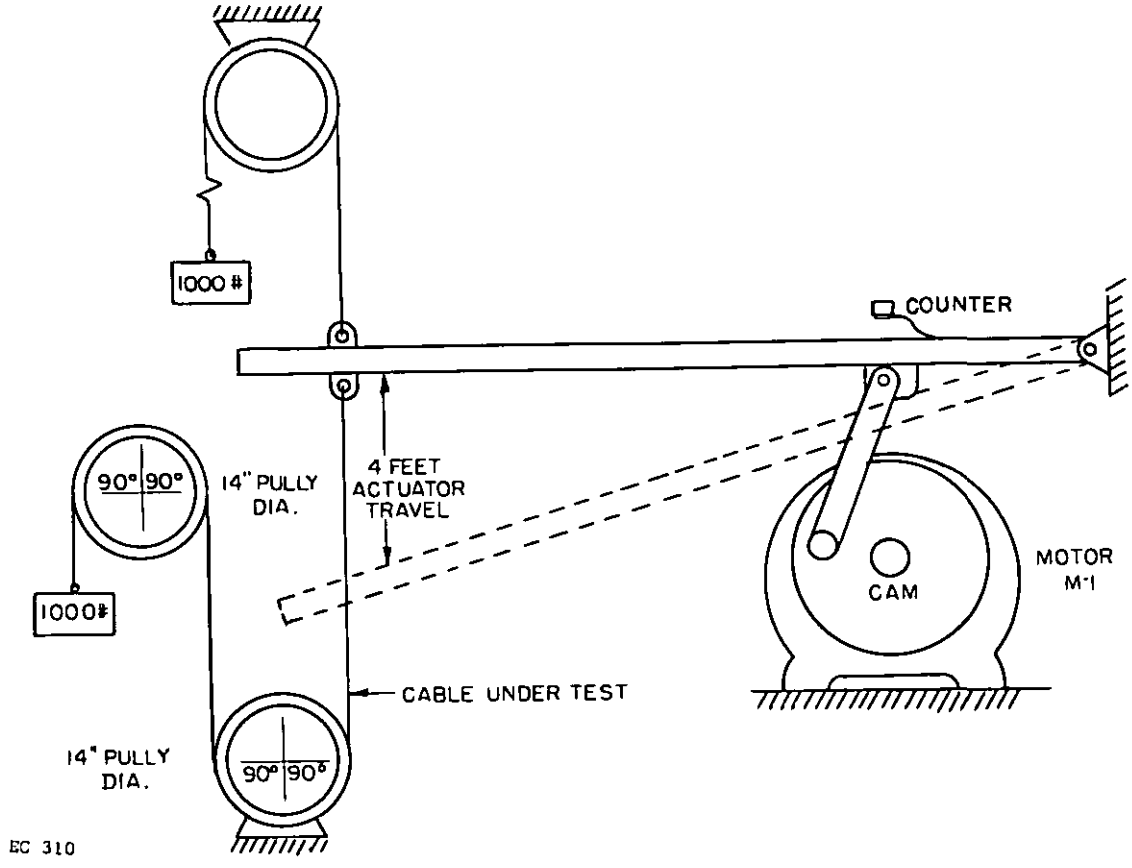


Figure 1. Cable, flexibility test equipment arrangement

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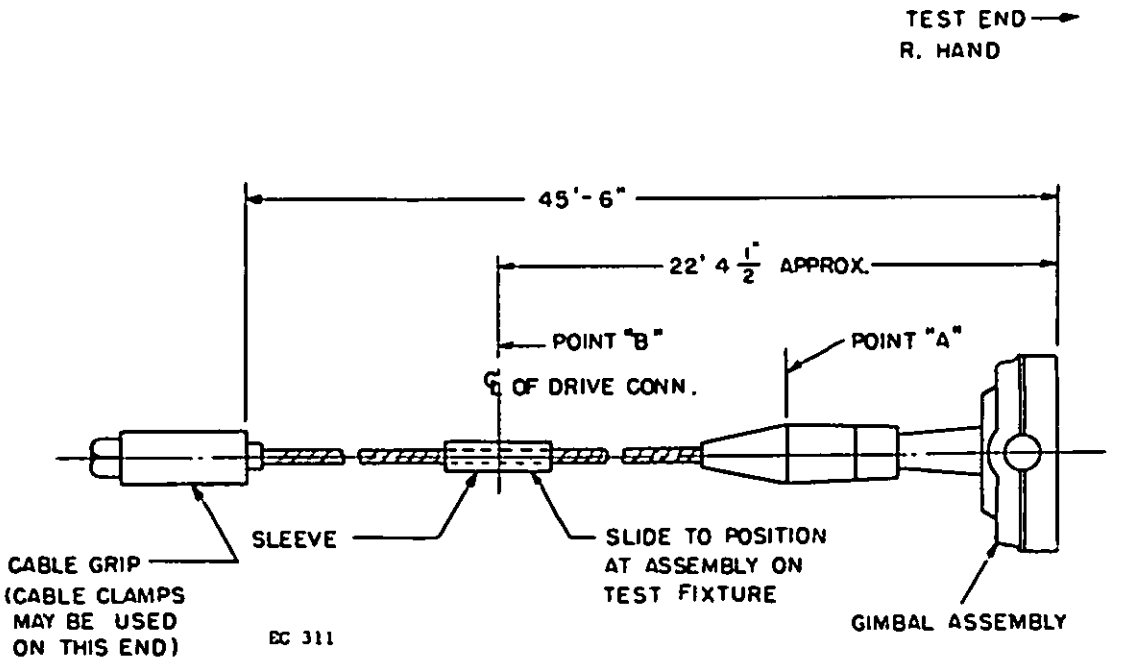


Figure 2. Test cable assembly

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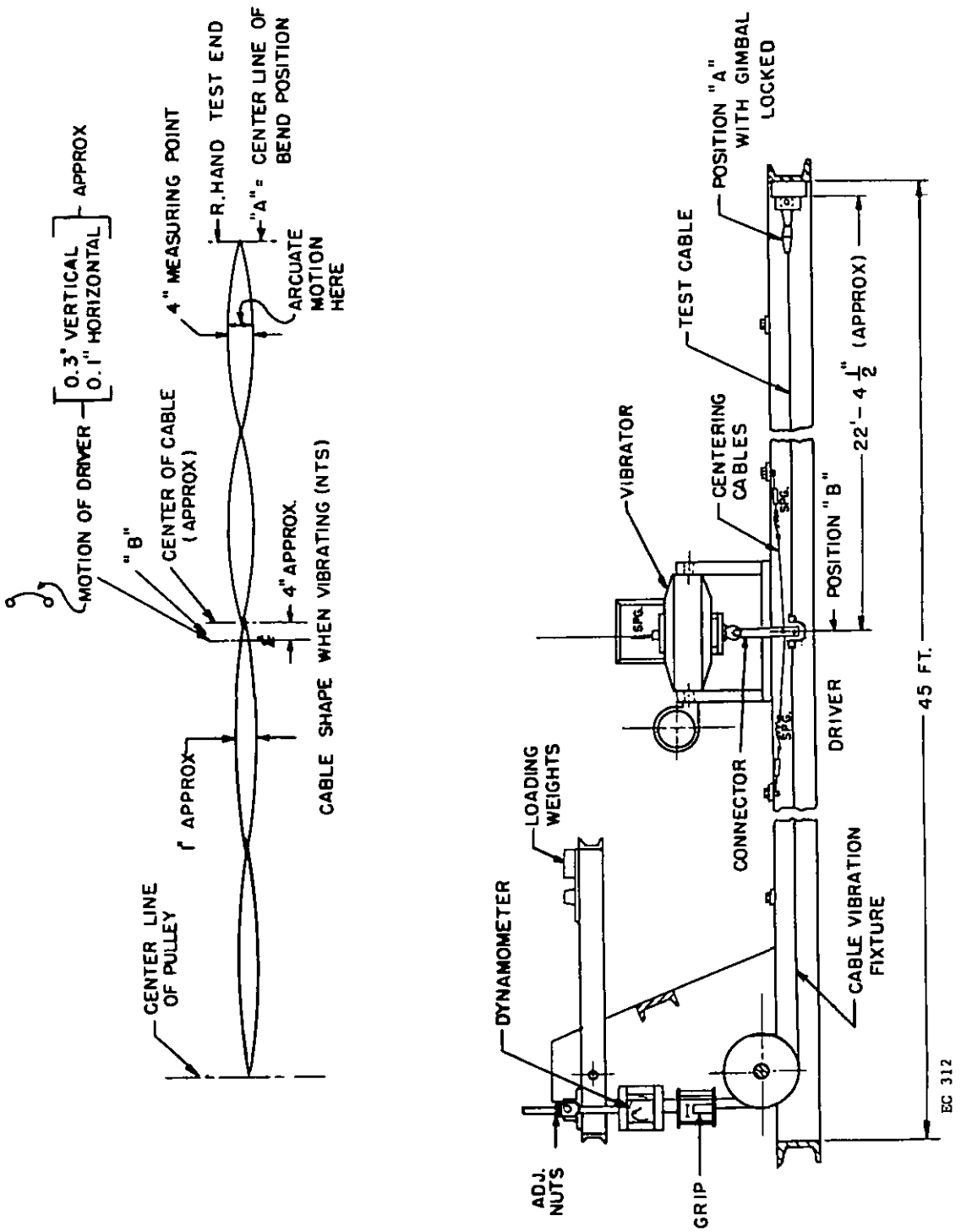


Figure 3. Cable vibration test

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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